

PROJECT REPORT:

Loan Approval Prediction using Random Forest

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1. Abstract

This project predicts loan approval (Y/N) using machine learning. Using Kaggle dataset (614 records), we cleaned data, handled missing values, and trained a Random Forest Classifier. The model achieved 75.6% accuracy.

2. Problem Statement

To automate the loan approval process for banks and reduce manual verification and risk of defaulters.

3. Dataset

Source: Kaggle. 13 columns including Gender, Married, Education, Income, Credit_History, Property_Area. Target: Loan_Status.

4. Methodology

Data Cleaning -> EDA -> Label Encoding -> Train-Test Split (80:20) -> Random Forest Model -> Evaluation.

5. Data Preprocessing

- Missing values filled with Mode (categorical) and Median (numerical).
- Dropped Loan_ID column.
- Converted all categorical data to numbers using LabelEncoder.

6. Model Used

Random Forest Classifier - Chosen because it prevents overfitting and gives high accuracy for classification.

7. Result

- Accuracy: 75.60%
- Confusion Matrix: $\begin{bmatrix} 18 & 25 \\ 5 & 75 \end{bmatrix}$
- Precision (Approved): 75%, Recall (Approved): 94%
- The model is excellent at identifying approved loans.

8. Conclusion

The model successfully predicts loan approval and can help banks make fast, data-driven decisions. In future, it can be deployed as a web app using Streamlit.

THANK YOU